

Mehraan Chowdhury

Sydney, AU • mehraan.chowdhury@gmail.com • 0416 810 665 • github.com/m-a-x-c
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PROFILE

- Master of Philosophy graduate from UWA with thesis on deep learning for medical screening, supported by 2 first-author IEEE conference publications
- Master of Information Technology (Artificial Intelligence) from University of Melbourne, WAM 81.1 (H1 / High Distinction average)
- 3.5 years' commercial front-end development experience building responsive cross-browser interfaces in HTML5, CSS3, JavaScript, and jQuery
- Proficient in Python, Java, C, and SQL with hands-on experience deploying deep learning pipelines using PyTorch, scikit-learn, and OpenCV
- Effective written and verbal communicator with experience presenting technical research at international IEEE conferences in the USA and New Zealand

EDUCATION

Master of Philosophy (Computer Science) Sep 2023 – Mar 2026

University of Western Australia

- Thesis: *Deep Learning-Based Screening of Obstructive Sleep Apnea Using 3D Facial Images*
- Ranked in the top 5% of theses examined, per examiner's report
- Two first-author IEEE conference publications produced from thesis research

Master of Information Technology (Artificial Intelligence) Mar 2021 – Dec 2022

University of Melbourne

- WAM: 81.1 (H1 / High Distinction average)
- Thesis: *Predicting Relevant Context for Programming Questions using Stack Overflow Answers*

Bachelor of Science (Neuroscience) Mar 2013 – Nov 2015

University of Melbourne

PUBLICATIONS

- M. Chowdhury, M. Bennamoun, F. Boussaid, S. M. Shamsul Islam, "Classifying Obstructive Sleep Apnea from 3D Facial Images Using Automated Landmark Detection and Geodesic Distances," *2025 IEEE Conference on Future Machine Learning and Data Science (FMLDS)*, Los Angeles, 2025, pp. 602–607. doi: 10.1109/FMLDS67896.2025.00098
- M. Chowdhury, M. Bennamoun, F. Boussaid, S. M. Shamsul Islam, "Automatic Detection of Peripheral Facial Landmarks Using 3D Facial Data," *2024 IVCNZ Conference on Image and Vision Computing New Zealand*, Christchurch, 2024, pp. 1–6. doi: 10.1109/IVCNZ64857.2024.10794203

PROFESSIONAL EXPERIENCE

Research Student (MPhil) Sep 2023 – Mar 2026

University of Western Australia, Perth

- Designed and implemented an end-to-end deep learning pipeline in Python (PyTorch) for screening obstructive sleep apnea from 3D facial scans
- Engineered software for automated 3D facial landmark detection, replacing manual annotation and improving reproducibility
- Built reproducible Linux-based workflows on GPU hardware, integrating Open3D, trimesh, MeshLab, and CloudCompare for 3D mesh processing
- Tracked code in Git and authored 2 first-author IEEE conference papers from the resulting research

Research Student (Master's Thesis) 2022

University of Melbourne, Melbourne

- Built an NLP pipeline in Python to predict relevant programming context from Stack Overflow answers using transformer models (Hugging Face) and NLTK

- Scraped and processed large volumes of developer Q&A data using BeautifulSoup and ChromeDriver
- Reported findings in a thesis contributing to a final WAM of 81.1

Front-End Web Developer

Apr 2016 – Dec 2019

Revo Interactive Ltd, Dhaka, Bangladesh

- Converted design wireframes into responsive production user interfaces using HTML5, CSS3, JavaScript, and jQuery
- Ensured cross-browser consistency across Chrome and Firefox
- Integrated SEO content into HTML markup to improve search visibility for client websites
- Diagnosed and resolved bugs across the front-end codebase, reducing time-to-fix on reported issues
- Contributed design and development ideas during sprint planning, helping shape feature direction across the team

TECHNICAL SKILLS

- **Languages:** Python, Java, C, JavaScript, HTML5, CSS3, SQL, Bash
- **Front-end:** jQuery, responsive design, cross-browser compatibility (Chrome, Firefox)
- **ML & Data:** PyTorch, scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Optuna, Hugging Face, NLTK, Jupyter
- **Computer Vision & 3D:** OpenCV, PIL, Open3D, trimesh, MeshLab, CloudCompare
- **Web automation:** BeautifulSoup, Selenium / ChromeDriver, Playwright
- **Tools & Methods:** Git, GitHub, Linux, LaTeX, Agile, Waterfall

EXTRACURRICULAR

- Member, Coders for Causes (UWA), 2024–2025 — attended club events and contributed to *Friends of the Grounds Tour Map*, a progressive web app for self-guided UWA campus flora tours.

REFEREES

Mohammed Bennamoun

Professor, Computer Science and Software Engineering

School of Physics, Mathematics and Computing

University of Western Australia

mohammed.bennamoun@uwa.edu.au

Farid Boussaid

Professor, Electrical, Electronic and Computer Engineering

School of Engineering

University of Western Australia

farid.boussaid@uwa.edu.au